AUTOFEAST AI — AUTONOMOUS FOOD ORDERING SYSTEM

Comprehensive Technical Inventory & System Architecture Specification

Academic Project: Final-Year B.Tech Project Report	**Target Domain:** AI Speech & Autonomous Food Logistics
Document Scope: Exact Technical Inventory & Verifiable Audit	**Status:** Production Deployed & Formally Verified
Live URL: autonomous-food-ordering-system.vercel.app	**Repository:** AutoFeast-Autonomous-Food-Ordering-System

Executive Architectural Overview

AutoFeast AI is a dual-tier autonomous food discovery, ordering, and recurrence-scheduling platform engineered with a resilient multi-layered Natural Language Processing (NLP) architecture. Designed specifically for Indian multilingual speech patterns (Tamil, English, Hindi, Tanglish, and Hinglish), the system integrates an on-device WebAssembly-quantized Whisper automatic speech recognition (ASR) engine for complete offline operation, backed by serverless Google Gemini 3.8 Flash Large Language Models for semantic disambiguation and zero-hallucination catalog mapping. This technical inventory details all exact model specifications, algorithmic pipelines, database schemas, cryptographic security gates, REST API contracts, and verified empirical test suites.

1. AI Model Names, Versions & Operational Parameters

Model Name	Exact Model ID	Quantization / Spec	SDK / Runtime	Primary Architectural Role
Whisper Multilingual Tiny	`onnx-community/whisper-tiny`	q8 (8-bit Quantized ONNX weights, 39 MB)	@huggingface/transformers v4.3.2 + ONNX Runtime WASM	On-device offline Speech-to-Text (STT) transcription in browser
Gemini 3.8 Flash	`gemini-3.8-flash`	FP16 Serverless Cloud LLM (Max tokens: 160-350)	@google/genai v2.4.0 (Node.js SDK)	Phonetic/semantic search fallback, Voice NLU intent parsing, Health guidance
Gemini 2.5 Flash TTS Preview	`gemini-2.5-flash-preview-tts`	Prebuilt Voice 'Kore' (24 kHz PCM / WAV base64)	@google/genai v2.4.0 (Node.js SDK)	High-fidelity natural server-side speech synthesis in EN, TA, HI

2. On-Device Whisper Model Specifications & Client-Side Runtime

Model Repository: `onnx-community/whisper-tiny` hosted via Hugging Face Hub.
Quantization Profile: 8-bit Integer Quantization (`dtype: 'q8'`) for encoder and decoder ONNX graphs, reducing model memory footprint while preserving cross-lingual phonetic accuracy.
Model Binary Size: Verified at **39.0 MB** (`const MODEL_SIZE_MB = 39` in `src/services/offlineSpeechEngine.ts`).
Client Runtime Engine: Client-side execution inside the user's browser using `@huggingface/transformers` v4.3.2 compiling to ONNX Runtime WebAssembly SIMD Multi-Threaded (`ort-wasm-simd-threaded.asyncify.wasm` and `ort-wasm-simd-threaded.asyncify.mjs`).
Audio Ingestion & Preprocessing: Captures raw microphone streams via `navigator.mediaDevices.getUserMedia({ audio: true })`. Employs a Web Audio API `AudioContext` resampling pipeline converting arbitrary hardware sample rates (44.1 kHz / 48 kHz) into **16,000 Hz single-channel (mono) Float32Array PCM**. Automatically prepends 160 ms (2,560 samples) of ambient silence and pads audio buffers to 30.0 seconds to stabilize Whisper's 80-channel log-mel spectrogram filterbank.
Decoding Hyperparameters: Configured with `task: 'transcribe'`, `chunk_length_s: 30`, deterministic greedy search `temperature: 0.0`, `repetition_penalty: 1.2`, and `no_repeat_ngram_size: 3` to completely prevent looping hallucination loops during repetitive Indic syllables.
Cache Storage & Explicit Consent Policy: Model weights, configuration graphs, and WASM binaries are stored in the browser's persistent Cache Storage API under `transformers-cache`. The system strictly enforces an explicit user consent gate (`downloadModelWithConsent`) and never initiates background model downloading over metered connections without user approval.

3. Google Gemini Model Configuration & Fallback Trigger Thresholds

Primary Model ID: `gemini-3.8-flash` (configured via server environment variable `GEMINI_MODEL` with default fallback in `api/search-fallback.js`).
API Endpoints: `POST /api/search-fallback`, `POST /api/voice/nlu`, `POST /api/voice/health-nlu`, and `POST /api/tts` (which utilizes `gemini-2.5-flash-preview-tts`).
Fallback Trigger Conditions: The Gemini search fallback is NOT invoked arbitrarily. It is governed by a deterministic dual-threshold evaluation in `src/context/AppContext.tsx` and `src/services/geminiSearchFallbackClient.ts`:

1. Zero Local Matches: Triggered when primary local fuzzy/phonetic search returns 0 matching catalog items (`result.matchedItems.length === 0`).

2. Low Confidence Threshold: Triggered when the top local search match has a confidence score strictly below **0.45** (`bestLocalScore < 0.45`).

3. Online Safeguard: The client inspects `navigator.onLine`. If offline, Gemini is entirely skipped (`isOffline: true`) with 0 ms network delay, immediately rendering local popular catalog suggestions.

4. Timeout Guard: An `AbortController` terminates serverless Gemini requests after an exact **5,000 ms** timeout.

5. Server-Side Rate Limiting & Caching: Protected by an in-memory client rate limiter (30 requests/minute/IP) and a 1-hour LRU cache (`CACHE_TTL_MS = 3600000`) capped at 100 entries.

4. NLP Algorithms & Fuzzy Matching Libraries

NLP / Matching Module	Implementation / Library	Algorithmic Technique & Mathematical Principles
Bitap Approximate Search	`Fuse.js v7.5.0`	Bitap algorithm with multi-key weighted scoring (name: 0.4, nativeNames: 0.3, tags: 0.3, category: 0.2, cuisine: 0.2), threshold: 0.4, distance: 100.
Damerau-Levenshtein Similarity	`FuzzyMatchEngine (src/services/fuzzyMatchService.ts)`	Dynamic programming matrix measuring edit distance including transpositions: Similarity = 1 - (dist / max(len1, len2)). High conf: >= 0.82, Med conf: >= 0.58, Low: < 0.58.
Token-Set Ratio (Word-Order Invariance)	`FuzzyMatchEngine (src/services/fuzzyMatchService.ts)`	Computes Jaccard intersection over union on normalized token sets: |tokens1 ∩ tokens2| / |tokens1 ∪ tokens2|. Handles inverted speech (e.g. 'chicken kothu parotta' vs 'kothu parotta chicken').
Phonetic Key Simplification	`FuzzyMatchEngine (src/services/fuzzyMatchService.ts)`	Unicode NFD normalization + diacritic removal. Collapses geminate consonants (iddli -> idli), maps Indic aspirated stops (th/dh -> t/d, ph/bh -> f/b, ee/oo -> i/u).
Cross-Script Phonetic Matcher	`CrossScriptPhoneticMatcher (src/services/crossScriptPhoneticMatcher.ts)`	Direct grapheme-to-phoneme mapping between Tamil Unicode (■-■), Devanagari (■-■), and Latin phonetic transliterations.
Canonical Food Family Table	`FoodFamilyTable (src/services/foodFamilyTable.ts)`	28 Canonical Indian dish families (Parotta, Dosa, Biryani, Idli, Tea, Pongal, Roti, Poori, Chaat, Paneer, etc.) with alias trees and generic-vs-specific variant tokenizers.
Indic Number & Conjunction Parser	`NlpParserService (src/services/nlpParserService.ts)`	Regex tokenizers extracting quantities in English (one..twenty), Tamil (■■■■■..■■■■■, ■■■■■■, ■■■■■), and Hindi (■■..■■) with conjunction splitting ('and', '■■■■■■■', 'aur', '+').
Restricted Food Interception Engine	`RestrictedFoodMatcher (src/services/restrictedFoodMatcher.ts)`	Strips generic culinary descriptors (crispy, hot, spicy, fresh, steamed in EN/TA/HI) to isolate core dish tokens before testing against nominee-restricted IDs.

5. Food Catalog Size, Schema & Multilingual Support

Catalog Size: Exactly **23 verified Indian dishes** curated in `src/data/indianFoodCatalog.ts`.
Supported Natural Languages:

- **English** (`en / en-IN`) — Full UI translations, TTS, ASR, and fuzzy matching.
- **Tamil** (`ta / ta-IN / ■■■■■`) — Native Tamil script + Tanglish transliterations across all dishes, categories, errors, and audio prompts.
- **Hindi** (`hi / hi-IN / ■■■■■`) — Devanagari script + Hinglish colloquial representations.

Closed Domain Food Schema: Each catalog item implements `FoodItem` with exact fields: `id`, `name`, `nativeNames { en, ta, hi }`, `description`, `basePrice`, `rating`, `cuisine`, `category`, `isVeg`, `image`, `restaurant`, `distanceKm`, `locations[]`, `platforms[]` (Swiggy, Zomato, AutoFeast with real-time fee & delivery time metrics), `tags[]`, `isDiabeticSafe`, `isLowSodium`, `isKidFriendly`, `calories`, `prepTime`.
Zero-Hallucination Policy: When feeding catalog items to Gemini in `api/search-fallback.js`, dishes are serialized into `COMPACT_MENU` with a strict whitelist check against `VALID_CATALOG_IDS`. Any item ID generated by Gemini not in this 23-item whitelist is unconditionally discarded.

6. Full Application Technology Stack & Infrastructure

Layer	Technology / Framework	Exact Version	Purpose in Architecture
Frontend Framework	React	`19.0.1`	Declarative component-driven user interface with hooks and Concurrent Mode.
Build Tool & Bundler	Vite	`6.2.3 / 6.4.3`	ESM-based HMR dev server and Rollup production bundling with WASM support.
Language	TypeScript	`5.8.2`	Strict static typing across client state, API contracts, and catalog schemas.
Styling Engine	Tailwind CSS	`4.1.14`	CSS utility-first styling configured with Warm Cream (#FFF8F0) & Tomato Red (#D93636).
Animation	Motion (Framer Motion)	`12.23.24`	Spring animations, micro-interactions, and modal fade/slide transitions.
UI Iconography	Lucide React	`0.546.0`	Accessible, tree-shakeable SVG UI icons.
Client-Side AI / ML	@huggingface/transformers	`4.3.2`	In-browser transformer inference running ONNX Runtime WASM.
Server Runtime	Node.js (ES Modules)	`v24.13.0 (dev) / Node 20.x`	Backend REST API server execution environment.
Backend Framework	Express.js	`4.21.2`	REST API routing, middleware, JSON body parsing, and route security.
Password Hashing	bcryptjs	`3.0.3`	Salted SHA/Blowfish password hashing with 10 salt rounds for nominee credentials.
Cloud AI SDK	@google/genai	`2.4.0`	Official Google GenAI SDK interfacing with Gemini 3.8 Flash & TTS models.
Database ODM	Mongoose	`9.10.3`	MongoDB object modeling, validation schemas, and cached global connection.
Primary Database	MongoDB Atlas	`v7.0 / v8.0 Cloud`	Production cloud document database for users, PIN hashes, nominees, and orders.
In-Memory Testing DB	mongodb-memory-server	`11.3.0`	Isolated, ephemeral MongoDB engine for automated unit and integration tests.
Email Notifications	Nodemailer	`10.0.14`	SMTP delivery for nominee OTPs, invite links, and real-time order approvals.
Deployment Platform	Vercel Cloud Platform	`Serverless v2`	Edge CDN static hosting + serverless function execution via vercel.json rewrites.

7. Voice Command Processing Workflow

Step 1 — Microphone Acquisition & Mutex Lock: User activates voice via header mic (■ `Voice Search`) or in-page voice button. `SpeechController.requestMic({ owner, language })` enforces a single-active microphone mutex, terminating any background TTS or competing recognition listeners.
Step 2 — Primary Cloud STT Execution: Web Speech API (`webkitSpeechRecognition`) initializes with language tag (`ta-IN`, `hi-IN`, or `en-IN`), capturing interim and final transcripts with up to 5 N-best candidate hypotheses.
Step 3 — Offline Whisper Fallback: If Web Speech emits `network` error, socket disconnect, or offline status, `SpeechController` automatically transitions to `OfflineSpeechEngine`. Audio PCM is buffered at 16 kHz mono, padded, and transcribed locally via Transformers.js ONNX WASM running Whisper Tiny (Q8).
Step 4 — Hallucination Cleansing: The raw transcript is processed through `cleanWhisperTranscript()`, stripping non-speech tokens (`[MUSIC]`, `[APPLAUSE]`), repetitive loops, and zero-confidence acoustic noise.
Step 5 — Two-Tier Natural Language Matching:
- *Family Matching:* `detectFoodFamilyIntent()` checks N-best candidates against 28 canonical Indian food families.
- *Fuzzy Scoring:* `FuzzyMatchEngine.matchCatalogFoodItem()` scores candidate strings against the catalog using Damerau-Levenshtein distance and Token-Set ratio.
- *Quantity Extraction:* `NlpParserService` extracts numerical and verbal quantity tokens (e.g. '2', '■■■■■■', '■■').

Step 6 — Action Dispatch & Visual/Spoken Feedback: Matched items are dispatched to the shopping cart, guided order modal, or scheduler, followed by spoken confirmation via `speechSynthesis`.

8. Gemini Fallback Workflow (Semantic & Acoustic Homophone Resolution)

Condition 1: Triggered when local catalog search returns 0 matching candidates or top candidate confidence score < 0.45.
Condition 2: If `navigator.onLine === false`, Gemini fallback is aborted immediately without network invocation, returning safe popular catalog recommendations.
Condition 3: If online, client invokes `POST /api/search-fallback` with payload `{ text, language }` bounded by a 5-second `AbortController`.
Server-Side Processing:
1. *Rate Limiting:* Validates request frequency (max 30 req/min/IP).
2. *Cache Check:* Evaluates in-memory LRU cache (`searchFallbackCache`).
3. *Acoustic Prompt Injection:* Supplies the 23-item `COMPACT_MENU` to `gemini-3.8-flash` with specific instructions to resolve acoustic homophones (such as Tamil acoustic misrecognition '■■■■■■' for '■■■■■■■' / Parotta, or '■■■' for '■■■■').
4. *Strict Output Validation:* `validateSearchFallbackResult()` parses Gemini's JSON array, discards any non-catalog IDs, drops confidence < 0.5, and caps results at 5 items.

Client Presentation: Validated items are returned with `isSmartSearch: true`, rendering a distinctive purple 'Smart search' badge.

9. Security PIN & Nominee Approval Workflow

A. Security PIN Verification Architecture:
• **4-Digit Numeric PIN:** User authentication and order confirmation require a verified 4-digit PIN.
• **Brute-Force Protection:** The server enforces a strict 4-attempt limit. On 4 consecutive failed attempts, the account is locked for **15 minutes** (`lockExpiresMs = Date.now() + 900000`).
• **Offline Disconnection Isolation:** `AuthService.verifyPIN()` explicitly detects network errors (`ERR_INTERNET_DISCONNECTED` / `isOffline`). Network connectivity failures do NOT increment failed PIN attempts and prompt the user to reconnect rather than penalizing attempts.
• **Single-Use Ephemeral Tokens:** Successful PIN verification issues a cryptographically signed HMAC SHA-256 token valid for **2 minutes** (`TOKEN_EXPIRY_MS = 120000`). Placing an order or schedule immediately consumes the token, preventing replay attacks.

B. Nominee Supervised Approval Architecture:
• **Restricted Food Enforcement:** Nominees can designate restricted dishes (e.g. high-sugar or allergen items) via `/api/nominee/restricted-items`.
• **Interception & Approval Escalation:** When a user attempts to order or schedule a restricted item, the order is intercepted. An approval request is recorded (`POST /api/nominee/approval-requests`) with an expiry of **10 minutes**.
• **Multi-Channel Notification:** The nominee is notified via Nodemailer SMTP email containing a secure token link to the Nominee Portal (`/nominee`).
• **Nominee Authorization:** Nominees authenticate with a dedicated Nominee PIN or password to approve or reject requests. Approved requests issue a single-use claim token allowing the user's order to proceed.

10. Complete REST API Endpoint Inventory

HTTP Method & Route	Auth / Guard	Functional Purpose & Contract
`GET /api/health`	Public	Database & server health check (returns database readyState, uptime, and memory).
`POST /api/tts`	Public / Cached	Generates natural speech audio via Gemini 2.5 Flash TTS in EN, TA, HI with server caching.
`POST /api/voice/nlu`	Rate Limited	Interprets complex voice turn intents, quantities, items, and location choices.
`POST /api/voice/health-nlu`	Rate Limited	Evaluates dietary and elder health constraints, suggesting mild catalog dishes.
`POST /api/search-fallback`	Rate Limited / Cached	Gemini 3.8 Flash search fallback resolving acoustic homophones against closed catalog.
`POST /api/auth/register`	Public	Registers user with name, email, phone, and optional 4-digit PIN hash.
`POST /api/auth/login`	Public	Authenticates user via email and password, issuing session JWT.
`POST /api/auth/pin-login`	Rate Limited (4 tries)	Authenticates elderly users via 4-digit numeric PIN.
`POST /api/auth/reset-pin`	Password Protected	Resets user PIN after verifying email and password credentials.
`POST /api/verify-pin`	Bearer Auth / 4-Attempt Gate	Verifies 4-digit PIN; returns single-use 2-minute order confirmation token.
`POST /api/nominee/invite`	Bearer Auth	Sends nominee invitation email with secure setup token via Nodemailer.
`POST /api/nominee/setup-credentials`	Invite Token	Initializes nominee account credentials and password.
`POST /api/nominee/login`	Public	Authenticates nominee into the Nominee Portal.
`POST /api/nominee/verify-pin`	Nominee Token	Verifies nominee-specific 4-digit PIN for high-privilege approvals.
`POST /api/nominee/set-pin`	Nominee Token	Configures or updates nominee 4-digit PIN.
`GET /api/nominee/restricted-items`	Bearer Auth	Fetches active dietary/allergen food item restrictions configured by nominee.
`POST /api/nominee/restricted-items`	Nominee Token	Adds a food item to user's restricted ordering blacklist.
`DELETE /api/nominee/restricted-items/:id`	Nominee Token	Removes a food item restriction from the blacklist.
`POST /api/nominee/approval-requests`	Bearer Auth	Creates a 10-minute pending approval request when restricted item is ordered.
`GET /api/nominee/approval-requests/:id/status`	Public / Token	Polls real-time approval status (pending, approved, rejected, expired).
`POST /api/nominee/approval-requests/:id/respond`	Nominee Token	Nominee submits approve/deny decision on intercepted order.
`POST /api/nominee/approvals/:id/consume`	Bearer Auth	Consumes approved request token to execute the order; prevents reuse.
`GET /api/cart`	User / Session Auth	Fetches active multi-restaurant shopping cart for user.
`POST /api/cart/add`	User / Session Auth	Adds food item to cart with quantity and source tag (voice, search, schedule).
`PUT /api/cart/update-qty`	User / Session Auth	Updates item quantity in cart.
`DELETE /api/cart/remove`	User / Session Auth	Removes specific item from active cart.
`DELETE /api/cart/clear`	User / Session Auth	Clears all items from cart.
`GET /api/schedules`	Bearer Auth	Fetches active automated meal recurrence schedules for user.
`POST /api/schedules`	Single-Use PIN Token	Creates recurring order schedule (breakfast, lunch, snacks, dinner).
`DELETE /api/schedules/:id`	Bearer Auth	Cancels and removes recurring schedule.
`POST /api/orders/create`	Single-Use PIN Token	Executes order, verifies single-use token, debits wallet, creates live tracking.
`GET /api/prices/live`	Public	Returns multi-platform price comparison across Swiggy, Zomato, and AutoFeast.

11. Testing Frameworks & Empirical Test Verification Results

Testing Frameworks: Vitest v5.0.3, JSDOM v30.1.1, React Testing Library (@testing-library/react v16.3.3), and mongodb-memory-server v11.3.0.
Empirical Test Verification Log: Executed on live codebase at November 2026.

Test Suite File	Scope & Validated Capabilities	Tests Executed	Status

`homepageAndHeaderLayout.test.tsx`	9-item header order, Voice Search button, profile dropdown items, 2-column dynamic search layout, 4 feature cards, food catalog.	7 tests	**PASSED (100%)**
`headerLayoutAndLocationVoice.test.tsx`	Header wrapping without clipping, accessibility toolbar options, location popover voice recognition and prompt triggers.	6 tests	**PASSED (100%)**
`geminiSearchFallback.test.ts`	Primary local vs Gemini fallback thresholds (< 0.45), zero-match fallback, offline suppression, 5-second timeout, Tamil homophone correction.	23 tests	**PASSED (100%)**
`voiceWorkflowAndSecuritySuite.test.ts`	Multi-dish voice parsing, 28 food families, quantity extraction, PIN lockout (4 attempts), network disconnect isolation (ERR_INTERNET_DISCONNECTED).	10 tests	**PASSED (100%)**
`server.mongo.test.ts`	Mongoose connection caching (global._mongoose), /api/health database readyState, credential sanitization (never logging passwords), fail-closed 503 auth routes.	6 tests	**PASSED (100%)**
Aggregate System Total	**Comprehensive end-to-end frontend, NLP, voice, security, and database integration tests.**	**52 tests**	**52 PASSED (0 FAILED)**

Report Prepared By: Antigravity AI Engineering Assistant
Verified Against Codebase: Git Commit `9c168aa` (Branch `main`)

B.Tech Project Evaluation:
Autonomous Food Ordering System with Multilingual Voice & Supervised Nominee Controls